

MONDAY

SEAPOW

@ SeaAirSpace
2023

Building a City

By ANN TROPEA, EDITOR-IN-CHIEF

Armchair experts and motivational speakers the world over are fond of saying that Rome wasn't built in a day. Less an historical account of Roman expediency, the saying is meant to convey the idea that great things take time. Civilizations don't happen overnight. And certainly, building a city represents a commitment to an idea.

The sprawling city you see before you here at Sea-Air-Space 2023 also wasn't built in a day. But close. It took three.

To read all of Seapower's coverage of Sea-Air-Space, visit our website, seapowermagazine.org.



Last Thursday, these exhibit halls were empty shells. Footsteps rang across football-field expanses, voices echoed, and dust motes — not bands — played in ballrooms. We got to work. Everyone, from Navy League staff to contractors and vendors, to exhibit builders, to operations and logistics experts rolled up their sleeves and began the process of turning nothing into something. And not just any something, but something amazing.

The exhibition booths themselves are an incredible feat on their own. They showcase possibilities, and in many cases, fantastic realities of technology and innovation that will bulwark our sea services and support their missions in



The Exhibit Hall under construction.

the years to come. The panel discussions and other events are built as well, and not only with the physical infrastructure of microphones, lights, speakers, and chairs. Our Navy League team built these events with dedication, with people, and with a commitment to an idea.

Our idea is this: What if we amass the best and brightest together for three days once a year,

put our collective heads together, and create real solutions that affect the viability of our seapower. What if we change our world. What if.

Welcome to our city. 🏡

APRIL 3, 2023

- 2 Small Businesses
- 4 Mental Health
- 6 Command Booth Presentations
- 7-10 Exhibit Hall
- 12 STEM Expo



NAVY LEAGUE of the UNITED STATES



Parker Ballam, "Professor Quark," inhales carbon dioxide as dry ice melts. (See STEM Expo photos on page 12).

Small Businesses Make Big Waves

The demand for small business innovation, technology, and solutions has never been higher

BY EDWARD LUNDQUIST, SENIOR CONTRIBUTOR
FOR SEAPOWER

If rural western Massachusetts looks like something out of a Norman Rockwell painting, it's because that's where Rockwell kept his studio. But in addition to the picturesque college towns, quaint villages, covered bridges, and magnificent scenery, the region also has a long history of heavy industry that dates to the industrial revolution. These industries historically harnessed rivers for power and created mill towns that made textiles, paper, leather goods, electrical components, automobiles, and guns.

Pittsfield, Massachusetts is located 40 miles from Albany, New York and 140 miles from Boston. With a population of about 44,000, it is the county seat of Berkshire County (pop. 129,000). For many years the town's business was defined by its largest employer, the General Electric Company, which manufactured transformers, electronics and plastics, and employed 10,000 workers. Like much of western Massachusetts' heavy industry, it has moved elsewhere.

When General Electric left, it took many good paying jobs with it. But today, General Dynamics Mission Systems (GDMS) has a large, state-of-the-art facility involved in the design and manufacturing of complex electronics for defense purposes, such as submarine combat systems. In fact, General Dynamics' business is growing, attracting new and highly skilled workers, and providing an economic engine for Pittsfield and its surrounding communities.

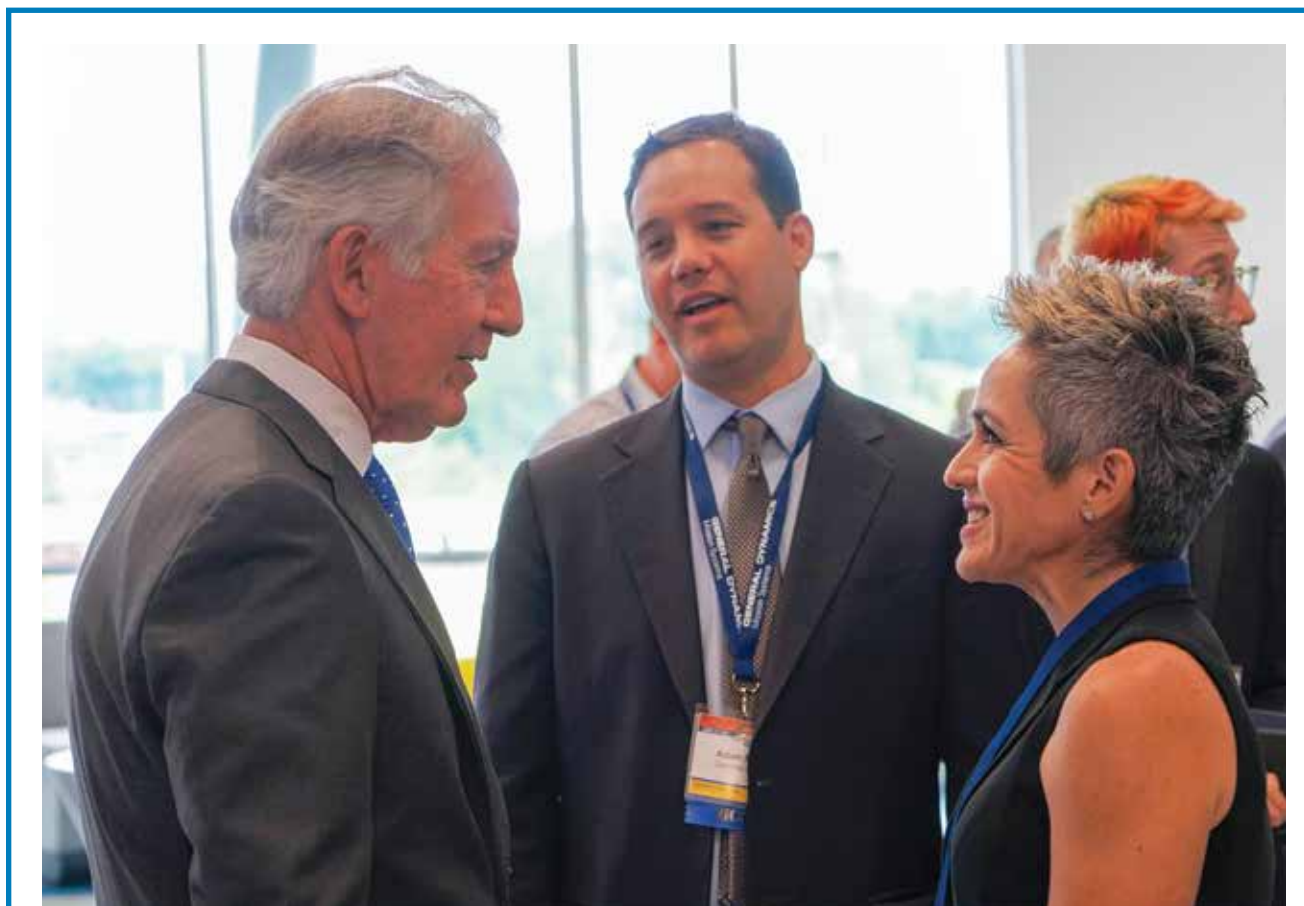
Global Threats Push Innovation

According to Ann Rusher, GDMS vice president of supply chain management, there have been unprecedented changes in the national security business, largely because, "China and Russia are introducing new technology and new capability at an alarming rate."

To counter that trend, Rusher said defense companies have had to accelerate the pace of innovation to work closely and more collaboratively with smaller businesses, including those that have not previously worked in the defense sector. "We need that innovative spirit and agility that small businesses can bring."

To accomplish that, GDMS is fostering better ties with the community and its supplier base. The company brought together vendors and partners — particularly small businesses — to the Berkshire Innovation Center (BIC) for a "Supplier Day."

Rusher said the event was aimed at reestablishing connections and relationships, and to make small companies aware of the resources in and around the Pittsfield area, as well as across the



Ann Rusher, vice president of supply chain at General Dynamics Mission Systems welcomes attendees at the Innovating for the Future event in June 2022.

country, to help them, "learn about, grow, and partner with us."

She said that big companies like General Dynamics truly rely on small companies, with their innovation and agility. Rusher said that today, more than 60% of GDMS current active supplier base are small businesses. "We added 104 small businesses to our active database just in 2021. And across all categories of small business, we've increased our spend over the last five years by over 15%. And every single one of the categories of small business — the HUBZone, service disabilities, veterans, and women owned — they've all been increasing, from five percent all the way to doubling."

Supply Chain Challenges

Rusher said General Dynamics not only wants to foster relationships between the company and small businesses, but also wants to facilitate the growth of those small businesses so that they can provide support to the entire defense industry. "By doing that, we can be a force multiplier for the government, and we can bring that innovative spirit, not just to us, but to the to the betterment of the country."

"We're a very successful company with an extremely talented workforce, but sometimes we need partners to help to solve some difficult problems," Rusher said. "Not only have these small companies helped us solve tough technical issues, but they often bring a technology or

a capability that's so unique and state-of-the-art that when combined with the mission knowledge that General Dynamics has, it really is the differentiator to solve a problem and deliver exceptional capability."

However, while the demand for innovative technology and solutions has never been higher, the number of small companies in the defense sector has declined precipitously.

According to Inside Cybersecurity, Deputy Assistant Secretary of Defense for Industrial

GDMS'S ANN RUSHER

"We added 104 small businesses to our active database just in 2021. And across all categories of small business, we've increased our spend over the last five years by over 15%."

Policy Jesse Salazar acknowledged that small businesses are under immense market pressures.

"The number of DIB (defense industrial base) small businesses has shrunk by more than 40 percent over the past decade," Salazar said.

»» SMALL BUSINESSES, Page 11



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Navy Prioritizes Mental Health with New Playbook

By VICKY UHLAND

The U.S. Navy has always been dedicated to ensuring that the bodies and minds of its Sailors are ready and prepared to win in combat. But there have been challenges over the years in helping Sailors with mental-health resources during active service, and as they transition out of the Navy into civilian life.

The Navy Culture and Resilience Office, N17, Office of the Chief of Naval Operations is addressing those challenges through its new Mental Health Playbook. This 28-page playbook, which was released in February, is designed to put mental health tools and resources into the hands of every Navy leader, no matter what their rank.

“Our goal is that everyone in our great Navy develops a shared understanding about how to conduct mental-health and preventative maintenance for our people, and then where to go for additional resources,” said Rear Admiral Brett Mietus, director of N17. “It’s an incredibly important topic to me and all of the Navy’s senior leadership.”

The Navy already offers a variety of mental-health resources. Navy N9 quality-of-life programs include Morale, Welfare and Recreation (MWR); Wounded Warrior; Mind Body



The Navy has a history of supporting mental health initiatives. Here, Navy Lt. Madeline Murphy, a physical therapist assigned to Navy Medicine Readiness and Training Command San Diego Directorate for Clinical Support Services, feeds a mini horse used for therapy in the hospital's courtyard in San Diego. Photo By: Navy Petty Officer 2nd Class Jake Greenberg

Mental Fitness (MBMF); and Sailor Assistance and Intercept for Life (SAIL). There are also family advocacy programs, childcare and relocation programs, assistance programs, non-medical counseling at fleet and family-support centers, and more.

The Navy League of the United States has mental-health resources as well. For example, Sea-Air-Space’s Transition Connection Job Fair, which will be held from 10 a.m. to 2 p.m. on Monday in the Cherry Blossom Ballroom, is

one of the ways the Navy League helps with transitioning Sailors’ overall well-being.

The Mental Health Playbook complements all these efforts through its mission to create a “climate of trust and respect with open, two-way communication; challenging inappropriate conduct or poor leadership; and eliminating stigma for seeking help.” The Playbook is organized into five sections:

Roles and Responsibilities, which is geared toward helping commanders create, “a community of support, where Sailors feel connected to the mission, the command and each other.”

Conversations That Matter, which provides strategies for conducting mental-health discussions.

Identifying and Responding to a Mental Health Related Concern, which discusses what to do when Sailors are in mental or emotional distress.

Navigating Support Systems, which helps Sailors find the right support at the right time.

Navy’s Mental Health Capabilities and Resources, which describes the clinical and non-clinical tools available both inside and outside the military and provides contact information for a variety of programs.

“Most of the resources that are in the playbook have been out there, but they just haven’t been put together in a way that’s easily digestible and then usable by a fleet leader,” Mietus said.

Mietus said the playbook is a response to requests and concerns from fleet members about the Navy’s approach to mental health. He noted that while older Sailors aren’t necessarily used to acknowledging or discussing mental-health issues, younger generations are much more attuned to their mental and emotional needs.

“Our goal is to eliminate stigma when it comes to mental-health care. I think the important thing for us all is to normalize conversations around it,” he said. 🍌

WELCOME FROM...

Navy League of the United States

On behalf of our 30,000 Navy League members, over 250 corporate partners, and supporters around the world, welcome to Sea-Air-Space 2023. Military and civilian leaders, industry and subject matters experts, exhibitors, and over 17,000 participants join us once again to be a part of something extraordinary.

I’m excited you’re here.

This year marks our 57th Sea-Air-Space Exposition, and it promises to be as bold as our theme: A Maritime Century. From protecting the borders of sovereign nations, to transatlantic security, to global shipping routes and trade, the security of our seas is more important than ever. Through an amazing feat of operations, logistics, and a year-long planning process, our team at the Navy League brings the universe of maritime possibilities and technologies to life.

I hope you make the most of Sea-Air-Space this year, and I warn you, there is a lot to see. But you certainly don’t want to miss the Sea Service Chiefs Leadership Panel on Monday morning, a live recording of the Maritime Nation podcast and remarks from our SECNAV, the Honorable Carlos Del Toro on Tuesday afternoon. Join us for a morning breakfast on Wednesday with Admiral Linda Fagan, Commandant of the Coast Guard. Feel free to stop any Navy League staffer and ask for their personal recommendations, as well.

Enjoy the show, and I look forward to seeing you back in 2024.

Mike Stevens

Chief Executive Officer, Navy League of the United States



CEO Mike Stevens, Retired Master Chief Petty Officer of the Navy

Q&A with Naval Supply Systems Command

Seapower magazine interviews Vice Commander Kurt J. Wendelken

What are the roles of the Naval Supply Systems Command?

NAVSUP and the Supply Corps conduct and enable supply chain, acquisition, operational logistics and Sailor & family care activities with our mission partners to generate readiness and sustain naval forces worldwide to prevent and decisively win wars.

How is the era of great power competition affecting NAVSUP's roles and responsibilities?

As Navy's end-to-end supply chain integrator, NAVSUP is working hard to maximize efficiencies and effectiveness in Navy's supply chains. One major initiative is Naval Sustainment System-Supply (NSS-S). Naval Sustainment System (NSS) is a combination of commercial best practices, process improvements, governance and oversight to maximize efficiencies and effectiveness within available means. NSS-Supply embraces industry best practices tailored for specific Navy needs and fleet operations in order to reduce maintenance turnaround times, increase end-to-end velocity of spares, repairs, and reduce costs.

Why does the Navy's supply chain need the transformation of the Naval Sustainment System-Supply?

NSS-Supply seeks to strengthen Navy's supply chain in order to increase readiness, thus

VICE COMMANDER KURT J. WENDELKEN

"We need to invest in sustainment up front, consider where prepositioning materials makes sense, and work better to reverse the current trends deteriorating material lead times."

enhancing combat capability and creating a model of sustainment that will allow the Navy to effectively generate readiness and sustain global navy power.

How does NAVSUP fit in the Navy's push for more distributed maritime operations and in the Marine Corps' expeditionary advance base operations?

Combined with the most robust, end-to-end logistics doctrine that the world has ever seen, NAVSUP provides uninterrupted (despite contested environment) supply chain management, bulk and aviation fueling capability, material handling equipment, contracting, hazardous material management, household goods and vehicle processing and postal operations to fleet, installation and other service components throughout every area of operations.



Kurt J. Wendelken

How will management of the supply chain be more integrated and streamlined with the warfighting commands?

By extension, NSS-Supply embraces industry best practices tailored for specific Navy needs and fleet operations in order to reduce maintenance turnaround times, increase end-to-end velocity of spares, repairs, and reduce costs.

What major segments of the supply chain are not owned by NAVSUP? How does the Defense Logistics Agency's roles compare with those of NAVSUP?

NAVSUP and the Supply Corps conduct and enable service specific supply chain, acquisition, operational logistics, and Sailor & family care activities with our mission partners to generate readiness and sustain naval forces worldwide to prevent and decisively win wars. The Defense Logistics Agency is the DoD executive agent for specific classes of supply that are common across the military services like food, fuel, construction material, and medical supplies. They also provide extensive warehousing services across DoD.

What are that current challenges that NAVSUP faces with the defense industry in executing the supply chain?

Executing an effective naval supply chain is increasingly challenging, in part due to the Defense Department's pressure on industry to become lean. We now face raw material shortages, weapon systems obsolescence, a shrinking skilled labor pool, excessive acquisition lead times, and a dwindling sub vendor base with a heavy reliance on sole source vendors. Maintaining our current warfighting edge requires better collaboration and transparency with industry. We need to invest in sustainment up front, consider where prepositioning materials makes sense, and work better to reverse the current trends deteriorating material lead times.

What new technologies look promising in aiding NAVSUP in streamlining the supply chain?

NAVSUP manages Navy's globally distributed, highly complex, and increasingly digital supply chains. LOG IT and Supply Chain systems are critical enablers in generating and sustaining readiness. NAVSUP is proactively taking steps to deliver modern digital solutions that support real-time operations to include system modernization, leveraging internet of things (IOT) technology and machine learning/artificial intelligence to improve asset visibility.

What can be done to expand competition among suppliers?

NAVSUP contracting offices attempt to enhance competition through early outreach in an effort to identify as many potential sources as possible. The use of Industry Days is also a popular method among NAVSUP contracting offices for specific types of procurements, such as ship repair or husbanding. These are established days where potential sources are invited to attend and learn as NAVSUP provides guidance on doing business with the government. This is a way to ease private sector concerns and generate interest.

What can the defense industry do to help the Navy improve its supply chain?

Just as the Navy has incorporated a "Get Real, Get Better" mantra, so we ask industry to embrace the same. We are all working towards a collective goal of supporting national defense, and it is imperative to operate with transparency and honesty, not distrust. Short-term, be accurate with contract schedules and deliver on time in accordance with contractual commitments. Increase the number and scope of strategic contracts to help offset material, labor, and financial stressors. Long-term, deepen the partnership with the Navy to create a more effective sustainment environment earlier in weapons system development. With more agility and cooperation, we can identify, plan, and overcome supply chain barriers to better support our fleet, which is the ultimate goal. 🏆

SEAPOW^{ER} @ SeaAirSpace 2023

Watch for our digital issues in your inbox on Tuesday and Wednesday mornings, featuring the latest news from Sea-Air-Space 2023 vendors and presenters.

Today's Command Booth Presentations



Naval Supply Systems Command (NAVSUP) Booth 301

9:30 a.m. – 10:00 a.m. ; 4:15 p.m. – 4:45 p.m.
Progress Update on NSS-Supply and P2P Logistics

CAPT Kelly House, USN, Director, Naval Sustainment System Supply, and Ms. Melissa Olson, Deputy Director, Naval Sustainment System – Supply

Insight, framework, and the impact of P2P-Log (Performance to Plan - Logistics), the analytical “Get Real” discovery, driving the “Get Better” of NSS-Supply, therefore improving our end-to-end supply chain.

11:45 a.m. – 12:15p.m.; 2:45 p.m. – 3:15 p.m.
Doing Business with NAVSUP

Mr. Chris Espenshade, Director, NAVSUP Office of Small Business Programs

This briefing will cover: service and supply contracting discussion points, how to best engage NAVSUP and specific manufacturing capabilities the Navy is seeking to further expand for weapon system support.

10:15 a.m. – 10:45 a.m.; 2:00 a.m. – 2:30 a.m.
Source Approval Process for Maritime/Aviation Platforms

Mr. Daniel Lenza, Director, Source Development Division, NAVSUP Weapon Systems Support

Lenza will cover the Navy's Source Approval Request (SAR) process for both Spare and Repair components in addition to referencing various business opportunity resources available for industry to leverage.

11:00 a.m. – 11:30 a.m.
Disruptions to the Supply Chain and Today's Challenges

Mr. Harold Valentine, Deputy Director, Plans and Operations and Enterprise Joint Lessons Learned System Manager, NAVSUP Headquarters

Valentine will provide insight into Supply Chain Challenges/Disruptions as it applies to plans and operations. Specifically, he will address the contested logistics ecosystem that consist of environmental, resourcing, national/global voice, political environment, access control, ability to defend, competition, and kinetic operational impacts. Valentine has almost 45 years of Naval experience as a Chief Petty Officer, Commander, and Senior GS Civilian Employee.

1:15 p.m. – 1:45 p.m.
Code 500 and Path Forward for Shipyards
CDR Abe Cook, USN, Assistant Supply Officer, Portsmouth Naval Shipyard

Portsmouth Naval Shipyard Supply Officer, CDR Abe Cook will be providing insight into the mission of shipyard Supply Departments, the unique challenges inherent in providing material support to shipyards and CNO repair availabilities, and how organizations can best align themselves to provide support to the Navy's industrial enterprise.

12:30 p.m. – 1:00 p.m.; 3:30 p.m. – 4:00 p.m.
Contracting

CAPT Josh Hill, USN, Director of Contracts, NAVSUP Weapon Systems Support and LCDR Jonathan Okonak, USN, Director, Acquisition Policy, NAVSUP Weapon Systems Support

In DoD contracting, time is of the essence and it's vital that our Vendor base understands the tertiary effects caused by delays in the various stages of the acquisition process. This briefing will cover: NAVSUP Weapon Systems Support contracting best practices, tools utilized to maximize support to the warfighter and methods to improve collab-

oration with our Industry partners. NAVSUP Weapon Systems Support is the “sinuate” connective tissue that bonds logistical solutions to fleet readiness, serves as the end-to-end supply chain integrator.

Coast Guard Booth 107

10:30 a.m. – 11:00 a.m.
USCG Risk Mitigation through Sustainable Aviation Fuel

Ms. Julie Berens, Energy Reliability Technical Warrant Holder, USCG, and Mr. Sam Alvord, Chief, Office of Energy Management, USCG

With the commandant's recently released Climate Framework directing the Coast Guard to adapt to the accelerating pace of change and take active steps to reduce the service's greenhouse gas emissions, the Office of Energy Management will summarize the data analysis, forecasted timelines and performance-based acquisition methodologies to facilitate this seamless transition in partnership with other government agencies.

11:00 a.m. - 11:30 a.m.
Oil Spill Response – Tech Efforts on the Horizon

Ms. Kirsten Trego, Deputy, Office of Marine Environmental Response Policy, USCG

As the federal on-scene coordinator for clean-up of oil discharges in the coastal zone, the Coast Guard and the broader National Response System must leverage advances in technology to effectively and efficiently detect, track, recover and remove oil pollution from the marine environment. From across the spectrum of emerging research and development to proven operational capabilities, this discussion will touch on how the Coast Guard is leveraging interagency, international and industry partnerships to drive advancements in oil spill response technologies.

11:30 a.m. – 12:00 p.m.
Evaluating Search Effectiveness: Keeping Pace with Technology?

CDR Matthew J. Mitchell, USCG, Chief, Office of Search and Rescue

Cmdr. Mitchell's discussion will center on the Coast Guard's need for a new method for reliably evaluating search effectiveness that meets the needs of the search and rescue mission.

1:30 p.m. - 2:00 p.m.
Interview with the USCG Deputy Commandant for Mission Support

VADM Paul F. Thomas, USCG, Deputy Commandant for Mission Support

As DCMS, VADM Thomas leads the 17,000-person organization that delivers the systems and people that enable the U.S. Coast Guard to perform its operational missions efficiently and effectively. During the interview, VADM Thomas will discuss the mission support enterprise and its current priorities and take questions from the audience.

2:00 p.m. - 2:30 p.m.
Robotic Process Automation

Dr. David F. Wiesenhahn, Modeling and Simulation Domain Lead, Office of Research, Development, Test & Evaluation and Innovation, USCG

The Coast Guard recently completed a research and development project to study the development, implementation and sustainment of robotic process automation (RPA). Applications examined during the research included automating data pulls from enterprise systems, streamlining procurement requests and other business processes, and uses within the areas of personnel, intelligence and logistics.

2:30 p.m. - 3:00 p.m.
Counter Unmanned Aircraft Systems
CAPT Eric M. Casper, USCG, Chief, Office of Specialized Capabilities

The Preventing Emerging Threats Act of 2018 grants the Department of Homeland Security (DHS) statutory

authority to counter credible threats from UAS to the safety or security of a covered facility or asset; this authority is paramount to DHS's mission to protect and secure the Homeland from evolving threats. Capt. Casper's briefing provides an overview of the scope of Coast Guard authorities, mission sets and actions that may be taken to support DHS, Coast Guard and Department of Defense counter-UAS authorized missions.

Naval Air Systems Command (NAVAIR) Booth 1747

9:30 a.m. - 10:15 a.m.
Small Unmanned Aerial Systems (SUAS) Portfolio Update

Col Victor Argobright, USMC, PMA-263 PM

Col Victor Argobright, Navy and Marine Corps Small Tactical Unmanned Aircraft Systems Program manager, will discuss program highlights, including a Maritime Search and Rescue demonstration at sea, two Medium Unmanned Logistics Systems-Air (ULS-A) Other Transaction Agreement/Authority (OTA) contract awards, operationally assessing the Tactical Resupply UAS and focus on expanding ISR Services Contractor-Owned, Contractor-Operated capability. Argobright will provide an update on where the program is right now and what to expect in the near future.

11 a.m. - 11:45 a.m.
Additive Manufacturing

Mr. Ted Gronda

NAVAIR Additive Manufacturing (AM) Program Manager, Mr. Theodore Gronda will discuss the recent deployment and operationalization of organic AM capability and how it is directly improving operational readiness by supplying direct operational support, addressing supply chain issues, providing maintenance efficiencies, reducing cost and improving part reliability through AM design.

1:30 p.m. - 2:15 p.m.
Program Executive Office: Unmanned Aviation and Strike Weapons

RDML Stephen Tedford, USN

RDML Stephen Tedford, program executive officer for PEO(U&W)—the central management authority responsible for the development, production, fielding and sustainment of the Navy's unmanned aircraft, weapons and target systems—will provide an overview of the U&W portfolio that includes nine programs and the Common Standards and Interoperability (CSI) office.

2:30 p.m. - 3:15 p.m.
Small Business Programs

Mr. Shelby Butler, Director, NAVAIR Office of Small Business Programs

SListeners will learn how to be prepared, what doors to knock on and what to consider when responding to opportunities. This panel will provide an eye-opening perspective on a firm's often overlooked and very underappreciated company profile, which is located on the Small Business Association's Dynamic Search website.

3:30 p.m. - 4:15 p.m.
History of Women in Naval Aviation

CAPT Erika Berry, USN; CDR Shannon Hoover, USN; LCDR Amanda Lippert, USN; CDR Jocelyn Liberg, USN; Moderated by Ms. Mikel Proulx

The panel will feature CDR Shannon Hoover, a P-8 test pilot and graduate of the U.S. Naval Test Pilot School and current Special Mission Aircraft IPT lead for the Maritime Patrol and Reconnaissance Aircraft (MPRA) Program Office; LCDR Amanda Lippert, a U.S. Naval Aerospace and Operational Physiologist; CDR Jocelyn Liberg, an F/A-18 pilot and the current deputy program manager of the Time Sensitive Strike Weapons for the Direct and Time Sensitive Strike Weapons Program Office; and CAPT Erika Berry.

Military Command Various Booth 349

Naval Information Forces (NAVIFOR) Chief of Naval Operations for Information Warfare (OPNAV N2N6)

U.S. Fleet Cyber Command / U.S. 10th Fleet (C10F)

Naval Information Warfare Systems Command (NAVWAR)

Naval Information Warfare Center (NIWC) Atlantic

NIWC Pacific

Program Executive Office (PEO) Digital and Enterprise Services (Digital)

PEO Manpower, Logistics and Business Solutions (MLB)

National Reconnaissance Office (NRO)

Naval Meteorology and Oceanography Command (CNMOC)

All Days, Full Exhibit Coverage
PEO MLB's MyNavy HR IT Solutions (PMW 240)

PEO MLB's MyNavy HR IT Solutions (PMW 240) Services creates and maintains mobile apps that enable Sailors to accelerate their careers through digital solutions. Stop by for live, interactive demos of the apps or find out how your app might be added to the Navy App Locker, which is home to all official Navy apps that touch the lives of every Sailor in the fleet. Learn about our projects in support of the Navy's HR Transformation efforts.

All Days, Full Exhibit Coverage
Naval Meteorology and Oceanography Command

Naval Oceanography gliders are autonomous underwater vehicles used by scientists and professionals (academia, oil and gas industry, military, etc.) around the world to collect a variety of oceanographic data in an effort to better understand ocean water columns in specific locations for different purposes.

All Days, Full Exhibit Coverage
Joint Tactical Networking Center

The Joint Tactical Networking Center (JTNC) supports the goal of rapid identification, characterization, procurement, fielding and sustainment of modular, innovative tactical communications products that ensure secure, interoperable, and resilient Command, Control, Communications, Computer, Intelligence, Surveillance, and Reconnaissance (C4ISR) capabilities.

Monday, Tuesday and Wednesday — See Booth for Schedule *Subject to Change
Information Warfare Engagement Zone

Various representatives from the participating commands will be available to engage with conference attendees on multiple topics and answer questions, so be sure to stop by and check the schedule. The information warfare community is comprised of approximately 63,000 professionals, including officers/enlisted, active/reserve, and civilians that are from the core specialties of intelligence and information professionals, cryptologic warfare, naval meteorology and oceanography, and the Navy's Space Cadre.

Monday, Tuesday and Wednesday — See Booth for Schedule *Subject to Change
Information Warfare Speaker Series

Come listen to senior leaders discuss the Navy's commitment to warfighting and the warfighter, now and in the future. Subject matter experts will be available to meet with attendees and highlight current technologies utilized by the fleet.

Company Name.....	Booths
(IDC) Isolation Dynamics Corp.	517
10X Engineered Materials LLC.	T519
3M Company	850

A-B-C

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Exhibit Hours

Prince George's Exhibit Hall:

April 3: 9 a.m. – 5 p.m.

April 4: 9 a.m. – 4:30 p.m.

April 5: 9 a.m. – 3:30 p.m.

Maryland Ballroom:

April 3: 8 a.m. – 4 p.m.

April 4: 8 a.m. – 4 p.m.

April 5: 8 a.m. – 11 a.m.

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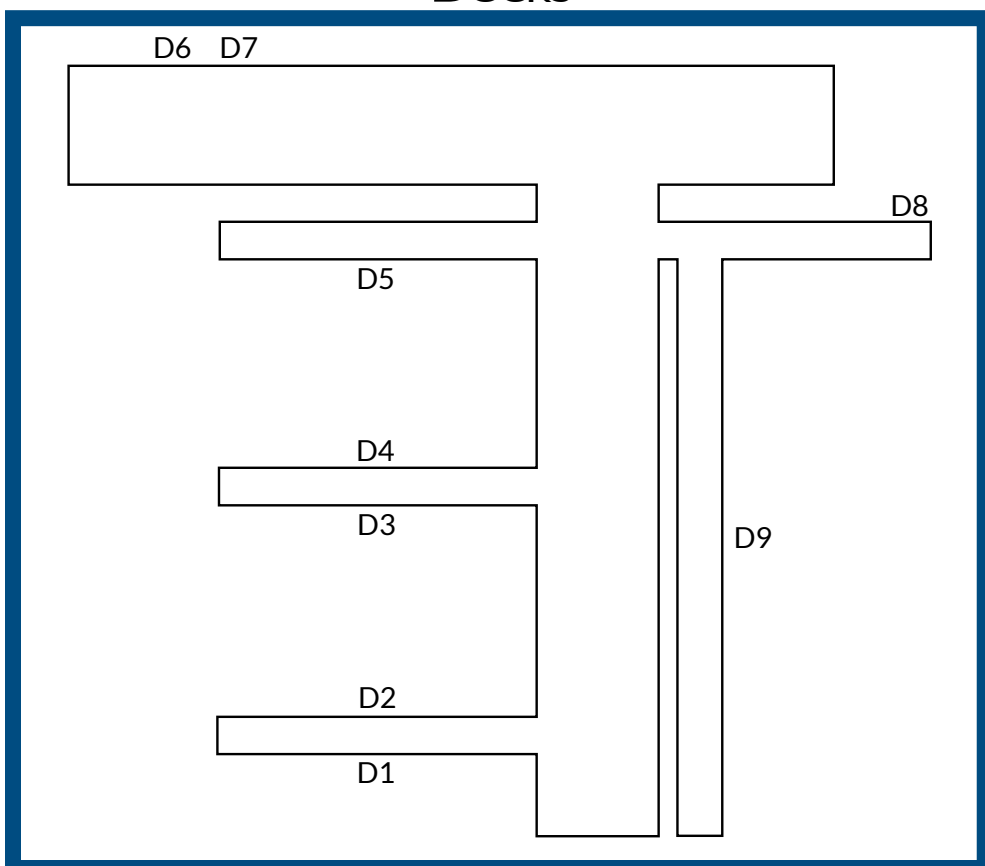
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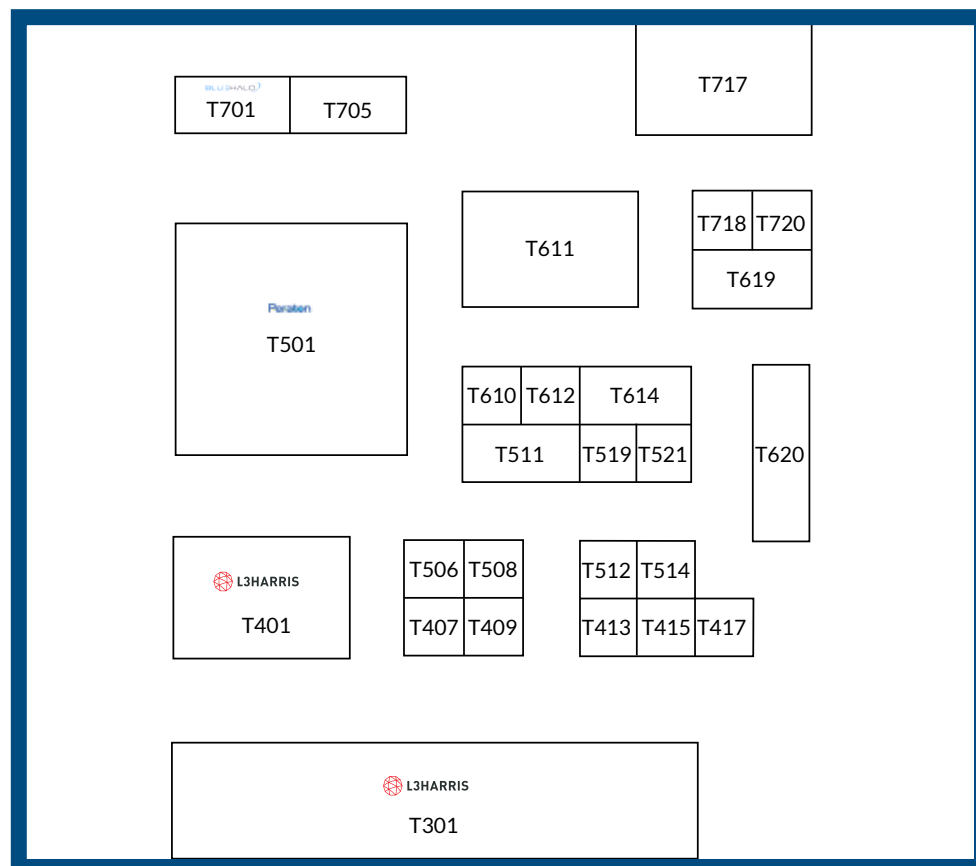
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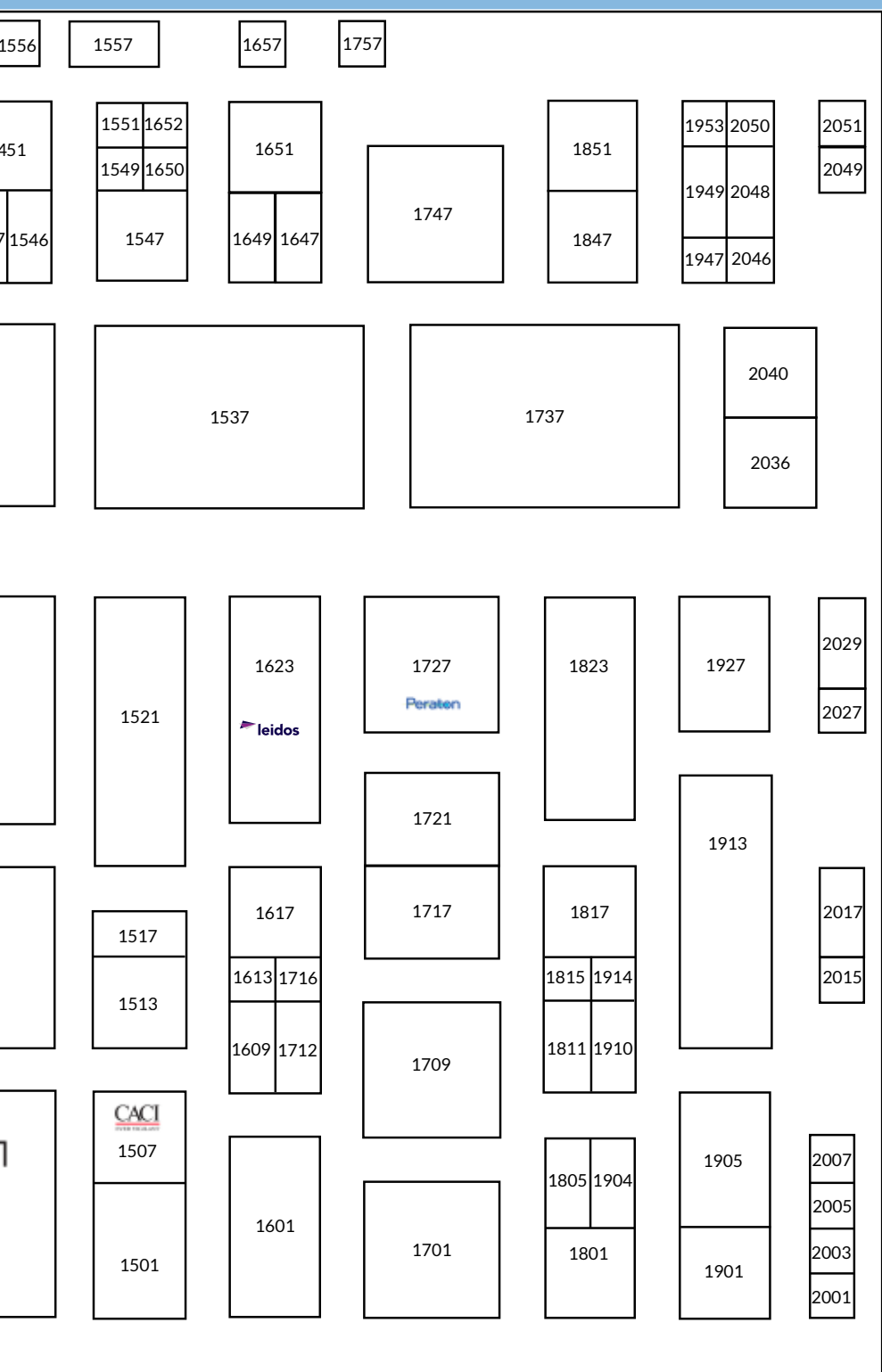


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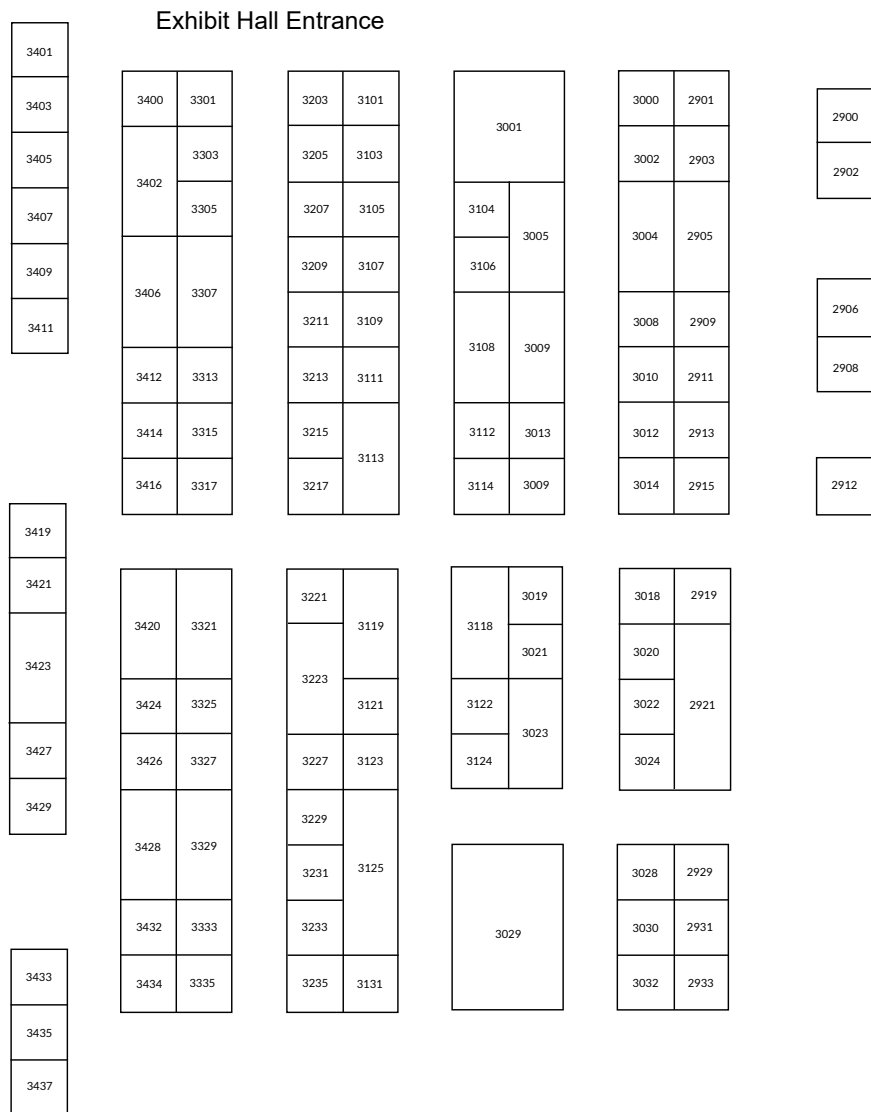
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April 3: 9 a.m. – 5 p.m.

April 4: 9 a.m. – 4:30 p.m.

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Maryland Ballroom:

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Small Businesses

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"One of seven believe they will never return to pre-pandemic levels of performance."

"Just when we need them the most, the supply chain is shrinking," Rusher said.

Rusher admits that it might be a little intimidating for a small company to establish a relationship and work with a large defense contractor like General Dynamics. "It might even be tempting to think that we like to go it alone. True, General Dynamics can do a lot of things. And we can do many of the things that perhaps a small business could do. But the reality is the small businesses we work with bring something very different, and way more than what we can do by ourselves," she said.

"We don't just want to work with you; we need to work with you," she said to the Supplier Day attendees.

BIC Innovation Hub

The BIC in Pittsfield is a multimillion-dollar collaborative initiative between the Commonwealth of Massachusetts, private industry, local colleges, and local government that opened in 2020. BIC serves as a confluence of technology and ideas. It offers conference rooms, offices and laboratory space, all designed to bring people together so they share knowledge and expertise to address challenges and seize opportunities.

"We offer world-class research and development facilities and equipment, interactive training and conference facilities, and shared access to advanced technology for local manufacturers," said Ben Sosne, BIC's executive director. "We can do more together."

Innovation centers like the BIC can offer online advanced manufacturing courses and access to content that serves companies and students, both locally and elsewhere. When potential employees learn about the technology being developed in the Pittsfield area, it can attract new qualified workers to companies like General Dynamics.

According to Sosne, a workforce with higher digital skills can command higher wages, but it also attracts more employers looking for people with those skills. "When we have a pipeline of new talent through apprenticeships, and the adoption of new technology, the higher the wages you can offer, and the more that you can attract that new talent. By working with local employers like General Dynamics and developing a curriculum that teaches methods and processes that meet their needs, you are essentially graduating an industry-ready group of engineering professionals and technologists."

"Employers like General Dynamics are an economic engine and a jobs-multiplier," said



Members of small businesses and prime contractors attend an SBIR break out session at the General Dynamics Mission Systems Innovating for the Future event in June 2022.

Benjamin Lamb Director of Economic Development with 1Berkshire. "One manufacturing job in the Berkshires supports 4.8 other jobs in the county."

The official regional economic development organization and regional tourism council of Berkshire County, 1Berkshire, represents the Berkshire business community and offers a powerful network of resources for members and companies. "This is where the synergy of marketing and economic development within the same organization in the same building with the same team can become very powerful," said Lamb.

Small Businesses are Essential Partners

"General Dynamics designs, engineers and makes all kinds of systems that are used for national security, so we take a lot of pride in that," said Pittsfield Mayor Linda Tyer. "They are an essential partner here in the city of Pittsfield and throughout the Berkshires. They are a large employer, providing 1,600 skilled jobs for our community. But there are many opportunities for our small companies to be part of the supply chain that serves General Dynamics, so that we are strengthening the economy here in Pittsfield and the Berkshires. Our plastics manufacturers and engineering companies help provide plenty of opportunity for those small businesses to benefit from the presence of General Dynamics."

Tyer said that workforce development is an essential part of the future and the success of General Dynamics and small businesses, no matter what kind of work they are doing. "It's incumbent upon academic institutions like our community colleges and four-year colleges, as well as institutions like the Berkshire Innovation Center (BIC), as well as the state agencies that provide workforce training opportunities and internship programs, to make sure they are partnering with each other and the employers who need talent and have the job opportunities," she said. "That's why having the BIC here is so important to the future of the innovation economy here in Pittsfield."

State Representative Tricia Farley-Bouvier, who represents Pittsfield, agrees. "General Dynamics reaches throughout the region to bring new talent to the Berkshires, not only with intellect and their skills, but the energy that they bring to our community. They spend their dollars in the outdoors and arts economy and in our shops and cafes, and volunteer in the Berkshire County. We want to ensure that we have a reliable local supply chain for this global company of General Dynamics, and ensure that the pipeline is a short one," she said. "I think workforce development is the biggest challenge right now, and that's across the board in every industry here in western Massachusetts. But we're very well positioned in Berkshire County because our high schools are producing really good workers and launching them into STEM careers, and that is an excellent source of talent for General Dynamics."

Farley-Bouvier cautioned that General Dynamics can't stand back and wait for the workforce to come to them. "General Dynamics has been and needs to continue to be part of that solution. They need to continue to be in at Taconic High School, MCLA (Massachusetts College of Liberal Arts), and Berkshire Community College and be part of those solutions. One of the most successful things that General Dynamics has done, and other smaller companies around Berkshire County have copied, is to provide paid internships. Paid internships are critically important because they level the playing field. It used to be that everybody took unpaid internships because there weren't a lot of jobs out there. But the only people who could take an unpaid internship were those students whose families could support them. The young people who were economically distressed had to take those low paying service jobs over the summer because they had to pay their bills," she said.

"Fortunately, General Dynamics is really invested in these students, and it's paid off for them," said Farley-Bouvier. "And we need to do a lot more of that to ensure that we have a reliable local supply chain for this global company of General Dynamics." 📌



HII CEO Chris Kastner looks on as Newport News shipbuilder employee Brain Treat demonstrates digital welding technology.



Helen Sullivan, 11, sets up tracks on a Gravitrax set at the STEM Expo.



Andrew Martinec, 10, tries on USMC armor and a helmet at the STEM Expo. His sister Camden Martinec, 9, and mom Jennifer Browning look on.



STEM Expo attendees explore the booths and experiments.

HII Dedicated to STEM

BY ANN TROPEA, EDITOR-IN-CHIEF

Amidst a maze of colorful exhibition booths, a bustling stream of kids and parents explore cutting edge tech, including 3D printers, tablet-controlled robots, rocket testing stations, and hands on (literally!) electricity experiments.

America's largest shipbuilder, HII (Huntington Ingalls Industries), is at the center of it all with a huge variety of interactive exhibit stations and lots of smiles from their enthusiastic staff. Grant Ronquillo and Kelsie Rountree (mechanical engineer and trade school coordinator for HII, respectively) both talked about the fun being at the Navy League STEM Expo, grinning as they watched kids learn how to weld in a virtual reality simulator.

President and CEO of HII, Chris Kastner, also spoke with *Seapower* about HII's dedication to STEM, including a rigorous internship program. "We have a good conversion rate from our interns, especially our technical interns, because we actually give them real work to do when they come into the company," he said. "They're not just doing busy work."

But ultimately, HII's focus is on the benefit to the larger community.

"If we don't invest our communities, in STEM, and make sure we have technical talent that can work in the shipyards, then we're not doing the right thing for the community," said Kastner.



Abbie Eastman gets her hair spiked by static electricity at the STEM expo.

The CMV-22B and Fleet Support

By Chris Misner, Senior Manager
Bell Military Sales & Strategy

It is axiomatic that robust logistics systems win wars on land and especially at sea. In a protracted naval battle, the importance of moving critical personnel, parts, munitions, medical supplies, and the world's most advanced fighters' powerplant around any Area of Responsibility (AoR) will become a limiting factor unless the Navy reassesses its approach to contested logistics and ultimately procures the equipment required to succeed.

With geopolitical tensions rising in multiple regions of the world, the U.S. military is closer than it has been in four decades to conflict with a peer/near-peer competitor. Increased capabilities demonstrated by potential adversaries are driving the need for rapid development of modern warfighting systems along with a renewed focus on industrial capacity and production. Pentagon planners must reconsider whether we have the capabilities to deter and/or defeat our adversaries and determine whether we have the industrial base in place to support protracted conflict.

If deterrence fails and conflict occurs, naval commanders will have to resupply their forces over vast distances, posing significant operational challenges where force structure does not match its strategic combat logistics requirements. The Bell Boeing CMV-22 delivers operational flexibility with the speed and range to move cargo and forces to support combat logistics for a distributed force in a contested environment across thousands of miles.

Unless action is taken soon, V-22 production will end when the logistic demand signal for potential major combat operations is reaching its peak. If the Navy is going to leverage the CMV-22 to solve complex logistics problems in contested environments and fully support

the Fleet, additional aircraft beyond the current Program of Record will be required. The Navy must recognize the reality that the V-22 production line will someday come to an end and leverage the opportunity to cost consciously add to the Program of Record to reap the benefits this aircraft can bring to fleet readiness and warfighting sustainment in contested maritime environments.

It isn't hard to imagine the need for an additional CMV-22 force structure that would be required in a protracted Pacific conflict. When you consider the amount of personnel and cargo that would be required to support such an effort, the problem of resupply is not just a Navy problem — it is a "Naval" & "Joint Force" problem.

In the case of an Indo-Pacific conflict, the Navy needs to be prepared for a scenario not seen in the past 80 years. The size of the Pacific AoR, and the potential logistical capability gaps that will undoubtedly come to light when executing Expeditionary Advanced Base Operations, Distributed Maritime Operations, and Littoral Operations in a Contested Environment, will drive Combatant and Naval Component Commanders to spend as much time planning how to support the force as they will employing it. The requirement to move passengers and cargo between

dispersed Expeditionary Advanced Bases, Forward Logistic Support Sites, and ships at sea will prioritize the need for a dedicated logistics connector. How the Navy and Joint Force leverages the versatility and capability of its aviation forces will be key — tiltrotor technology will play an essential role.

The speed, range, endurance, and flexibility the V-22 has demonstrated over decades of performance, including combat, paved the way for the Navy to develop its own variant of the Osprey. Tiltrotor technology opens doors to new missions and operational possibilities that provide the Navy with the overmatch and operational flexibility it needs to solve the problems of contested logistics and provide support to the Fleet when operating as a distributed force. Just as the Marine Corps and Air Force found new and creative ways to leverage tiltrotor technology, the Navy will also learn how to operate and integrate the Osprey into its Carrier and surface support groups of the future.

The Navy must have an updated, survivable, and sustainable concept of operations for combat logistics in the Pacific. The Osprey is unique in it can support and sustain the Fleet and Joint/Combined Force while simultaneously augmenting the Carrier Air Wing's combat capability and lethality. Bell Boeing has decades of experience producing Ospreys. That expertise will be required to continue to sustain and upgrade the V-22 for several decades to come. Procuring additional Ospreys and protecting the nation's only tiltrotor industrial base is critical to ensure the Navy and the Joint Force are ready to meet the required logistical throughput envisioned in a future era of extended combat both in the littorals and at sea.

The appearance of U.S. Department of Defense (DoD) visual information does not imply or constitute DoD endorsement.



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CREDIT: DVIDS

Sikorsky Delivers 5,000th “Hawk” Helicopter



PHILIPPINE SEA (Oct. 27, 2022) An MH-60R Sea Hawk, attached to the “Saberhawks” of Helicopter Maritime Strike Squadron (HSM) 77, takes off from the flight deck of the U.S. Navy’s only forward-deployed aircraft carrier, USS Ronald Reagan (CVN 76), in the Philippine Sea. HSM 77’s missions include surface warfare, anti-submarine warfare, and a variety of support roles for Ronald Reagan and Carrier Strike Group (CSG) 5. Ronald Reagan, the flagship of CSG 5, provides a combat-ready force that protects and defends the United States, and supports alliances, partnerships and collective maritime interests in the Indo-Pacific region. (U.S. Navy photo by Mass Communication Specialist 3rd Class Eric Stanton)

BY RICHARD BURGESS, SENIOR EDITOR

Earlier this year, Sikorsky marked a milestone with the delivery of the 5,000th “Hawk” variant helicopter, a UH-60M Black Hawk for the U.S. Army. The Hawk family of helicopters has built a legacy of accomplishment and will continue to support medium-lift, maritime combat, utility, rescue, and special operations require-

ments for the U.S. military and international operators for decades into the future.

“Sikorsky is committed to all H-60 and S-70 operators and look forward to enduring partnerships on the Hawk program in the years to come,” said Nathalie Previte, the company’s vice president of Army and Air Force Systems. “It remains our priority to fulfill our customers’

requirements and meet the needs of current and future operational units. With unmatched versatility and global interoperability, Hawk aircraft continue to advance 21st century security,” she added.

Sikorsky’s highly skilled and experienced workforce manufactures the latest generation of Hawk aircraft built in Stratford, Connecticut, and the international S-70 variant is built at Lockheed Martin’s PZL Mielic facility in Poland. Customers worldwide depend on the Black Hawk platform and its derivatives. As of this

Voices of the Sea Podcast

from the Navy League of the United States

Episode 1 | Listen now

SIKORSKY’S NATHALIE PREVITE

“It remains our priority to fulfill our customers’ requirements and meet the needs of current and future operational units. With unmatched versatility and global interoperability, Hawk aircraft continue to advance 21st century security.”

writing, U.S. Navy Seahawks and U.S. Army Black Hawks are providing disaster relief to earthquake-stricken communities in Türkiye.

The U.S. Army is the largest Black Hawk operator, for which the helicopter continues to provide support during military and civil operations around the world, in addition to its significant contribution towards Joint All Domain Operations. As U.S. Army Future Vertical Lift aircraft are fielded, the Black Hawk will remain the foundational tactical air assault and utility aircraft for the U.S. Army.

“2023 is a momentous year for Sikorsky as we celebrate our 100th anniversary. Sikorsky, as a company, has been forged by the Black Hawk,” said Sikorsky President Paul Lemmo. “The Black Hawk and its variants deliver when reliability and performance are nonnegotiable. Hawk aircraft continue to demonstrate their versatility.”

Insitu's Integrator UAV Goes VTOL to Increase Versatility, Reduce Footprint

BY RICHARD R. BURGESS, SENIOR EDITOR

Insitu, a Boeing Company, has developed a vertical takeoff and landing (VTOL) capability for its Integrator Group 3 unmanned aerial vehicle (UAV), a capability which will increase the launch and recovery flexibility of the Integrator and reduce the footprint of the required ground support equipment.

The Integrator VTOL, as the system was recently named, combines an eight-rotor lifter UAV called FLARES III that carries aloft the stock Integrator and releases it for an airborne launch. For recovery at the end of the mission, the same lifter carries a cable aloft to snag the hooked wingtip of the Integrator, which is then lowered to the deck.

Justin Pearce, vice president for Engineering at Insitu, said in an interview with Seapower that the concept was developed beginning in the mid-2010s for the company's widely used, smaller ScanEagle UAV under the FLARES II (Flying Launch and Recovery System II) project, using a smaller rotary-wing UAV as the lifter. The FLARES II can lift 55-60 pounds.

In July 2022, the FLARES III was tested on board a U.S. Navy guided-missile destroyer, where it successfully launched and recovered an Integrator.

"We've had a number of discussions [with the Navy] following that successful demo in July with what are the next steps regarding their MSAR CONOP [concept of operations]," Pearce said. "MSAR is the unclassified name for the program."

"We've been maturing the end-to-end system solution we would offer to bring the full capability forward," he said. "So really, this tactical demonstration in July was then trying to understand what I'll call state of the industry in terms of VTOL capability."

For the larger Integrator UAV, the two-stage VTOL concept initially named FLARES III features for the VTOL component a UAV with four arms, each with two rotors. It is built by Hood Technology Mechanical, also known as HoodTech. Insitu matured the FLARES III in the background while the FLARES II was demonstrated.

The battery-powered lifter UAV can carry more than 165 pounds aloft with a full fuel load, enough to lift and launch an Integrator. Pearce said that no modifications to the Integrator are needed for the lifter UAV to grasp it and launch it, a feature that offers numerous advantages.

The use of the lifter UAV eliminates the need for the pneumatic launch rail for the Integrator. The lifter UAV also carries aloft the vertical ca-



PHOTO: Insitu

ble needed to snag and recover the Integrator, eliminating the need for the SkyHook vertical tower that normally captures the Integrator. The reduced footprint of the VTOL system conserves valuable deck space on a ship.

The lifter UAV also offers the advantage of a more stabilized launch and recovery.

"It is rock-solid in the sky," Pearce said. "Why? Because it's not running red line on [its] motors, rotors and batteries. That's one of the challenges that everybody has with hybrid VTOL. You naturally want to skip on the battery and the sizing of the components you put on board. That decoupling of the decision for how to size your VTOL subsystem from the airframe allows us to [increase] the resilience of the VTOL system to wind gusts. It allows us to have a wider operating limit in terms of sea state."

Pearce said the VTOL system keeps station over a moving vessel and moves with it and can hover and recover the Integrator higher up than with a SkyHook or net. It can be offset from the landing spot, or approach from any direction.

"Integrator does not change its recovery behavior," he said.

Pearce said the Integrator VTOL has the advantages of a fixed-wing UAV and the advantages of a VTOL launch and recovery. The VTOL operation also reduces the wear and tear on the Integrator.

"It retains payload capacity, airworthiness certification of the base platform because you're not changing it," Pearce said. "MIL standard,

cyber, that's all retained. It's all about mobility. No need to tow trailers. Gives a truly expeditionary capability. It starts to rival Group 4 [UAV] capability."

"We see it today as no-compromise VTOL," he said.

The U.S. Navy operates several RQ-21A Blackjack systems based on the Integrator. More than 30 different payloads have been integrated with the Blackjack. The Marine Corps operated the RQ-21A for several years but has placed them in storage in favor of the MQ-9A Reaper as part of its Force Design 2030 concept.

Pearce sees a potential for future RQ-21A service in the Marine Corps particularly in expeditionary advance base operations.

"I see this as an asset that can fill a lower echelon need, where your Group 4's aren't going to get pulled off from the teams that need them," he said. "They should consider how this could apply to their CONOPS. This certainly should create a new look for Congress as well as the services in terms of how they can employ their RQ-21 investment."

Insitu also sees similar potential with the ongoing operations of the ScanEagle UAVs from Coast Guard cutters.

"FLARES II is something we would like them to adopt," Pearce said. "We understand their increased need for deck space, increasing the amount of hangar space they need for the good work they do getting drugs off the sea. So, we're in constant communication about how we can crawl that space back." 🦾

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